Teaching Statement

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Teaching Philosophy

My research combines two ways of understanding collective decisions: proving what voting rules guarantee, and measuring how they behave on real and synthetic data. I want my students to learn both, and to see how each informs the other. A theorem tells them what a rule guarantees; an experiment shows them how often that guarantee matters in practice. Three principles follow.

- *Every concept is paired with data.* When I introduce a definition, students also apply it to a concrete instance. In collective decision making this is natural: students can load a real participatory budget from Pabulib, compute the outcomes of several voting rules, and only then study the axioms that explain why the outcomes differ.
- *Students learn to ask questions, not only to answer them.* Alongside standard exercises, I will set open-ended problems, such as finding the statistical model of votes that best describes a city's ballots. Such problems have no answer in the textbook: students must design a method, test it, and defend their conclusions.
- *Computational practice is taught explicitly.* Writing clean, reproducible experimental code is a core skill for computer scientists, yet it is rarely taught as such. I will teach it with `prefsampling` and `mapof`, the Python libraries I released, so that students work with research-grade tools from the start.

Teaching and Mentoring Experience

At Jagiellonian University (2018–2021), I taught *Introduction to Programming* (a full-year course), *Logic and Set Theory*, and *Linear Algebra* (one semester each) to groups of about fifteen undergraduate students. For my research community, I co-authored the *Guide to Numerical Experiments on Elections* [C15] and released `prefsampling` and `mapof`, and I co-presented the tutorial *Experiments in Social Choice* at IJCAI 2023, AAMAS 2023, and AAAI 2024. I have given more than thirty talks.

I have co-supervised more than ten BSc and MSc students, and work with students led to conference papers at AAAI 2025 [C19] and ECAI 2025 [C22]. I start each student with a well-defined question for which data and tools are at hand, so that a first result comes quickly, and then move to an open problem whose answer I do not know. I review experimental design, code, and writing as carefully as proofs. As President of the PhD Students' Association at Jagiellonian University, I led an elected body representing doctoral students.

Teaching at EPFL

I am prepared to teach core bachelor courses (introductory programming, discrete mathematics, algorithms) and master courses in artificial intelligence and algorithmic game theory. In large bachelor courses, I would apply the same principles at scale: weekly programming and problem-solving exercises with quick feedback, automated where possible, and short problems during lectures that show me which ideas students have not yet understood. Exams would test both proofs and computation. I would also like to develop two new courses:

- *Algorithms for Collective Decisions* (master): voting, participatory budgeting, fair division, and matching, with real data throughout and a final project on a real participatory budget.
- *AI and Democracy* (master seminar): AI-mediated deliberation, AI-written summaries of public consultations, and AI agents that vote on people's behalf, open to students from computer science and the social sciences.

Beyond the classroom, I would like to offer student projects with Swiss municipalities that run participatory budgets.